CFD Analysis of Fluid Flow in Pipe with Various Roughness

By:- **KISLAY ANAND**
(23BMV1073)

Introduction

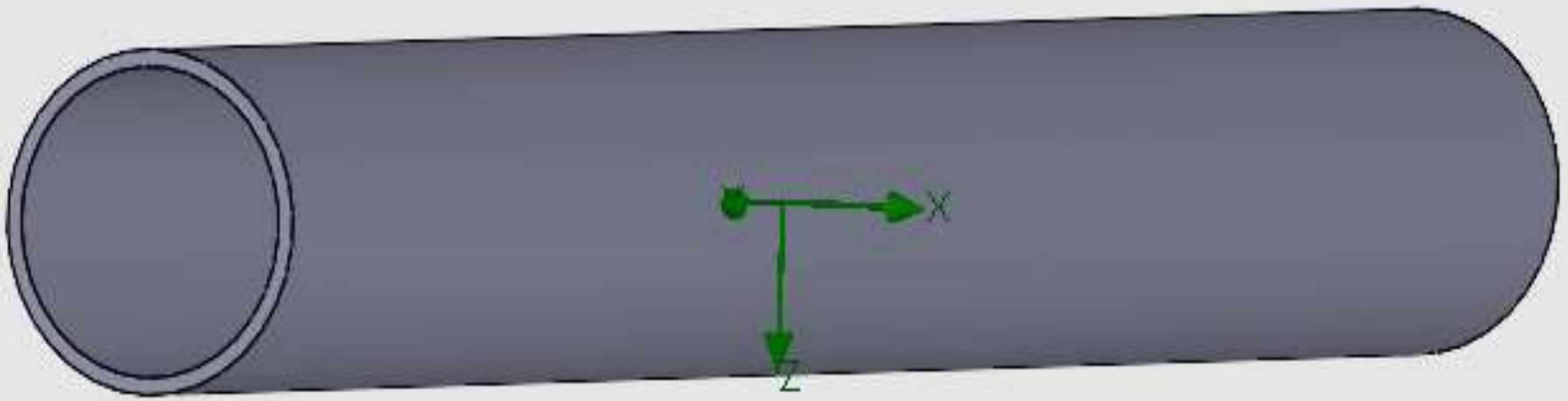
- Internal pipe flow is a fundamental topic in fluid mechanics and is widely encountered in engineering systems such as water supply pipelines, oil and gas transportation, heat exchangers, and HVAC systems. The performance and efficiency of these systems depend significantly on the behavior of fluid flow inside the pipes.
- In real-world applications, pipe interiors are rarely perfectly smooth. Surface roughness caused by corrosion, scaling, erosion, or manufacturing imperfections increases friction between the fluid and pipe walls. This results in higher pressure losses and energy consumption.
- Computational Fluid Dynamics (CFD) provides a powerful tool to analyze internal pipe flow and understand the influence of surface roughness on flow characteristics such as velocity distribution and pressure loss.

Project Overview

- The objective of this project is to investigate the effect of surface roughness on internal pipe flow using CFD simulation.
- The study focuses on comparing three different cases:
  - Smooth pipe flow
  - Low rough pipe flow
  - Rough pipe flow
- The simulations are performed using **SolidWorks Flow Simulation**. By analyzing pressure loss, velocity distribution, and temperature, the project aims to understand how surface roughness influences flow behavior in pipelines.

Literature Review

- Previous studies have examined the effects of surface roughness on pipe flow through experimental and numerical methods.Nikuradse (1933) conducted classical experiments on turbulent pipe flow using artificially roughened pipes and demonstrated that surface roughness significantly affects friction factor and pressure loss.
- Moody (1944) developed the Moody diagram, which relates Reynolds number, friction factor, and relative roughness for pipe flow.Later studies by Versteeg and Malalasekera (2007) showed that CFD techniques can accurately simulate internal flow behavior and provide detailed visualization of velocity and pressure fields.
- These studies establish that roughness increases



Pipe Geometry

- The pipe geometry was created in SolidWorks with the following dimensions:

Pipe length = **500 mm**
Outer diameter = **50 mm**
Wall thickness = **2.5 mm**
Inner diameter = **45 mm**

- The inner diameter represents the flow domain where the fluid moves through the pipe.

- The geometry is designed to ensure sufficient flow development along the pipe length.

Simulation Setup

- The CFD simulation was performed using **SolidWorks Flow Simulation.**
- Fluid selected: **Water**
- Boundary conditions applied:
- Inlet condition -> Velocity inlet
  Inlet velocity -> **0.2 m/s**
- Outlet condition -> Pressure outlet
  Outlet pressure -> **0 Pa**
- Wall condition -> No-slip boundary condition
- Flow regime considered -> **Turbulent**
- A computational mesh was generated with refinement near the pipe wall to capture velocity gradients accurately.

Roughness Cases

- Three different surface conditions were analyzed:

Case	Roughness
Smooth pipe	0 µm
Low rough pipe	50 µm
Rough pipe	200 µm

- Reynolds Number Calculation:

Formula: **$Re = (\rho V D)/\mu$**

oFor water:

$\rho = 998\ kg/m^3$
$V = 0.2\ m/s$
$D = 0.045\ m$
$\mu = 0.001\ Pa \cdot s$

❑Re≈8982 > 4000; Hence, the flow is proved to be turbulent.

Moody's Chart

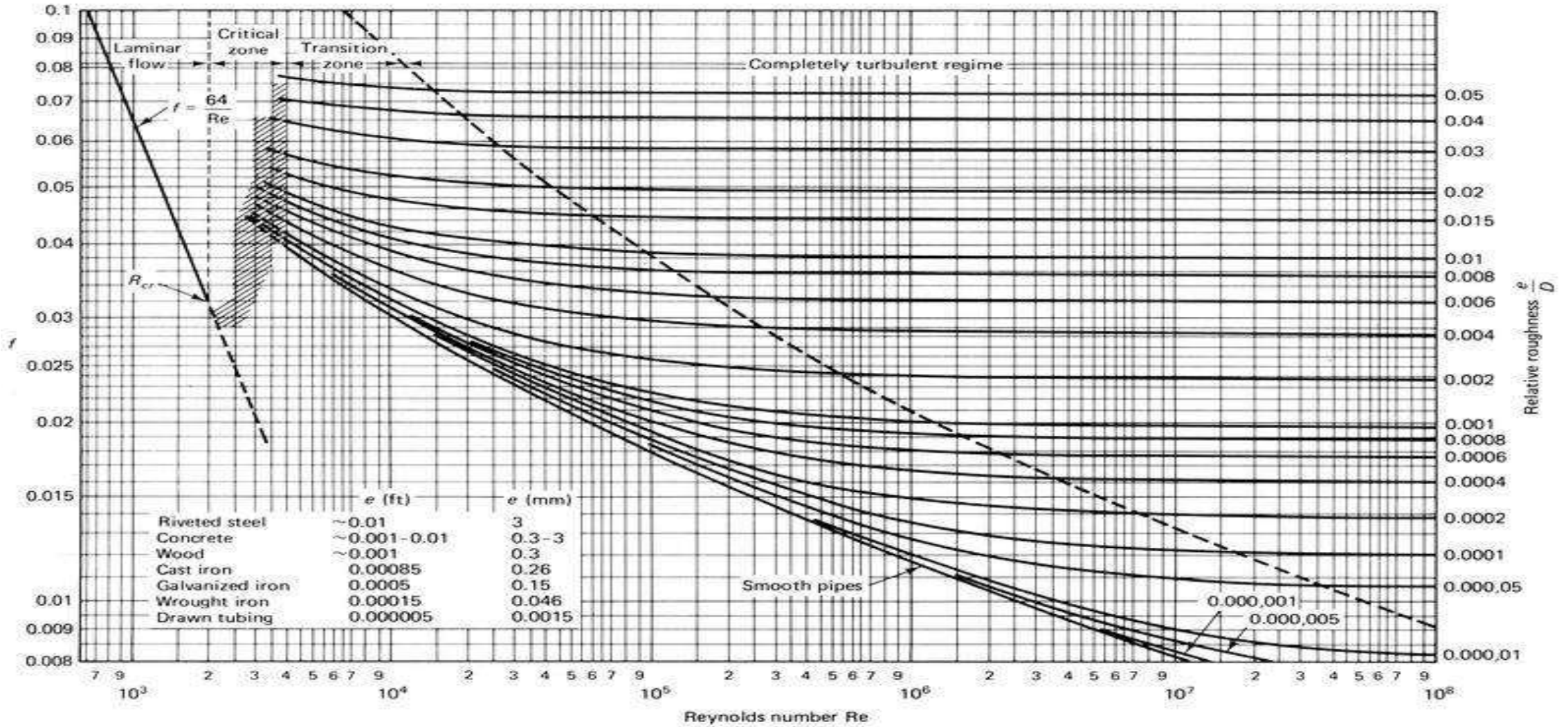


Figure 7.13 Moody diagram. (From L. F. Moody, *Trans. ASME*, Vol. 66, 1944.)

Darcy-Weisbach Equation

- $\Delta P = f \times (L/D) \times (\rho (V)^2)/2$

Surface roughness affects the friction factor, which in turn influences pressure loss in the pipe.

ΔP = pressure loss
f = friction factor
L = pipe length
D = pipe diameter
ρ = fluid density
V = velocity

From Reynolds number and Moody's chart, we get friction factor for each of roughness value.

Friction factor

Pipe Type	Roughness (μm)	Friction factor, f
Smooth pipe	0 μm	0.032
Low rough pipe	50 μm	0.035
Rough pipe	200 μm	0.042

Wall Shear Stress Relation

- Wall shear stress represents the frictional force exerted by the fluid on the pipe wall.
- It can be related to pressure drop using:

$$\tau = \frac{\Delta P\, D}{4L}$$

- Where:
- τ = wall shear stress
  ΔP = pressure drop
  D = pipe diameter
  L = pipe length
- Higher roughness increases wall shear stress and results in higher energy losses.

Pressure loss at each surface roughness

Pipe Type	**Roughness (µm)**	**Friction factor, f**	**Pressure loss, ΔP, Pa**
Smooth pipe	0 µm	0.032	7.1
Low rough pipe	50 µm	0.035	7.8
Rough pipe	200 µm	0.042	9.31

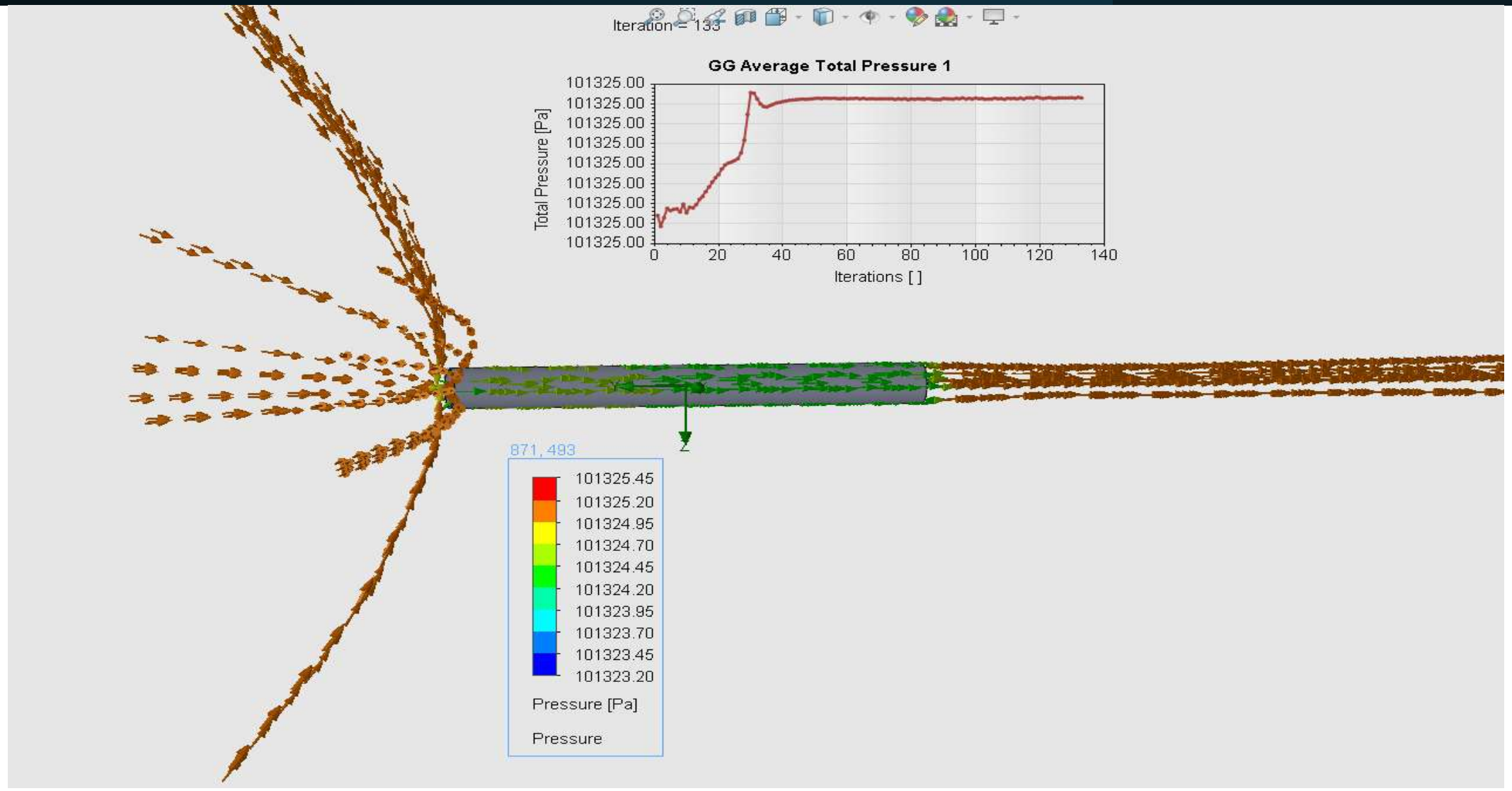
Wall Shear Stress at each surface roughness

Pipe Type	Roughness (µm)	Friction factor, f	Wall Shear Stress, τ , Pa
Smooth pipe	0 µm	0.032	0.16
Low rough pipe	50 µm	0.035	0.18
Rough pipe	200 µm	0.042	0.21

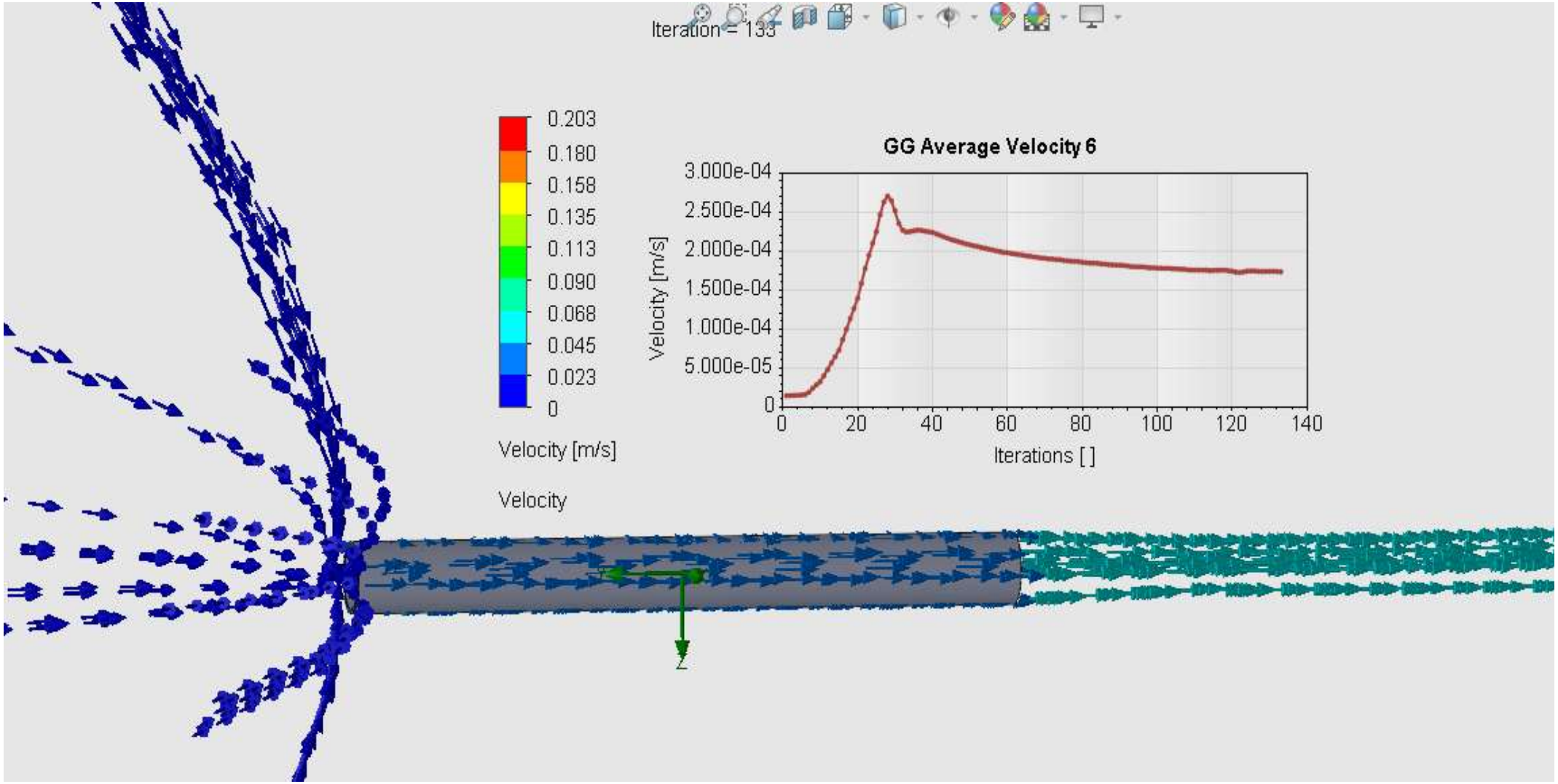
Preliminary Observation

- The smooth pipe case was simulated first to establish a baseline reference.
- The simulation results showed:
- • A stable velocity distribution along the pipe
  - Pressure drop of approximately **7.1 Pa**
  - Fully developed flow along the pipe length
- These results are consistent with theoretical predictions from the Darcy–Weisbach equation.

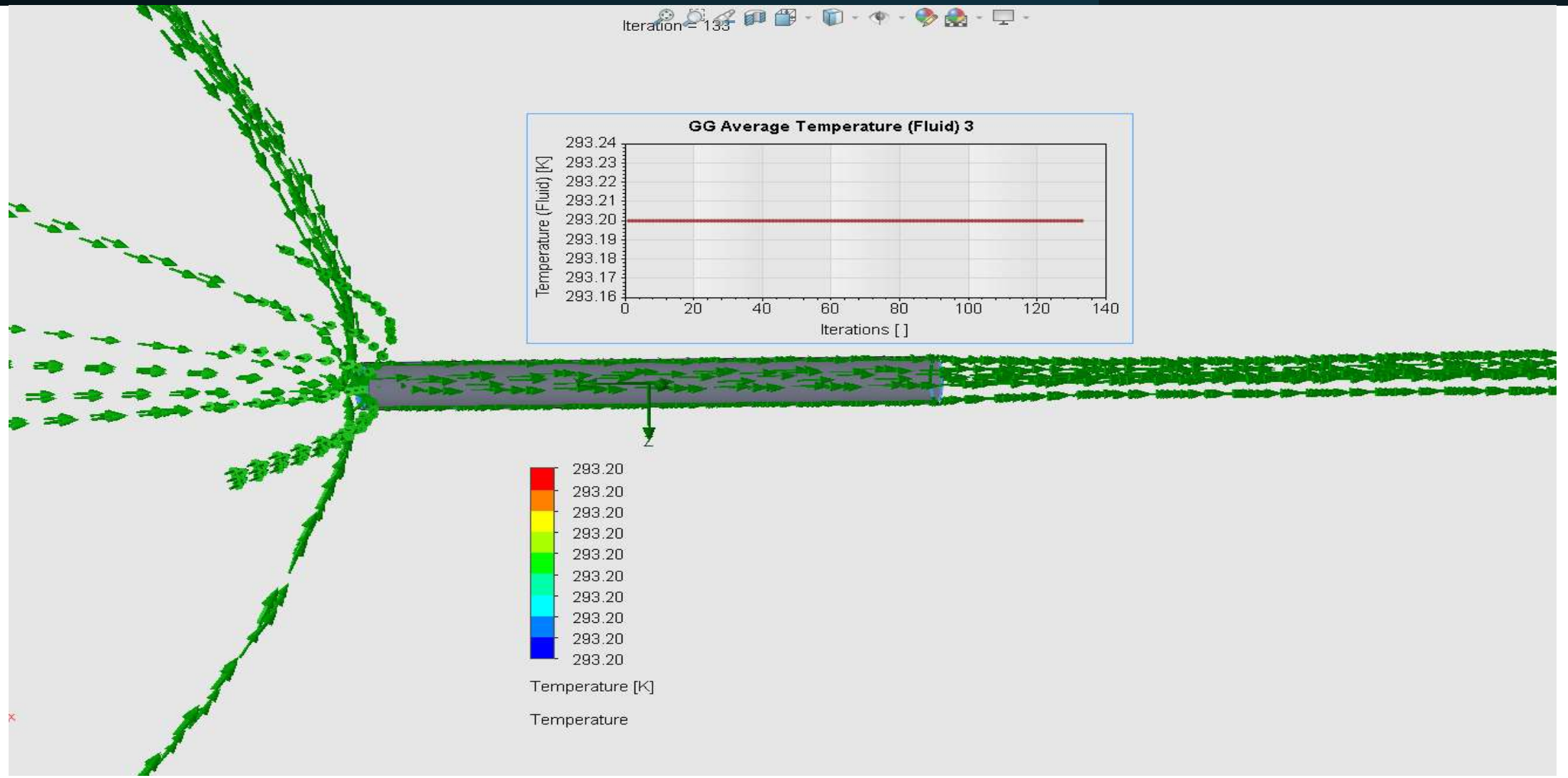
Results for the smooth pipe :
Pressure



Velocity:


Iteration = 133
0.203
0.180
0.158
0.135
0.113
0.090
0.068
0.045
0.023
0
Velocity [m/s]
Velocity
GG Average Velocity 6
3.000e-04
2.500e-04
2.000e-04
1.500e-04
1.000e-04
5.000e-05
0
Velocity [m/s]
0
20
40
60
80
100
120
140
Iterations [ ]

Temperature:



Expected Results

- Based on fluid mechanics theory and previous studies, the following trends are expected:
  - ❖ Smooth pipe
    → Lowest pressure loss
  - ❖Low rough pipe
    → Moderate pressure loss
  - ❖Rough pipe
    → Highest pressure loss

As surface roughness increases, the friction factor increases, resulting in higher pressure drop and higher wall shear stress.

Conclusion

- This project investigates the influence of surface roughness on internal pipe flow using CFD simulation.
- The pipe geometry and simulation setup have been successfully developed, and the smooth pipe case has been analyzed as a baseline reference.
- Further simulations for low rough and rough pipes will allow comparison of pressure loss and flow characteristics.
- The study helps validate classical fluid mechanics concepts and demonstrates the capability of CFD tools in analyzing pipe flow behavior.

References

- Çengel, Y. A., & Cimbala, J. M. (2018). *Fluid mechanics: Fundamentals and applications* (4th ed.). McGraw-Hill Education.
- Moody, L. F. (1944). Friction factors for pipe flow. *Transactions of the ASME, 66*(8), 671–684.
- Nikuradse, J. (1933). *Laws of flow in rough pipes* (NACA Technical Memorandum No. 1292). National Advisory Committee for Aeronautics.
- Schlichting, H., & Gersten, K. (2017). *Boundary-layer theory* (9th ed.). Springer.
- Versteeg, H. K., & Malalasekera, W. (2007). *An introduction to computational fluid dynamics: The finite volume method* (2nd ed.). Pearson Education.